Technical specifications for Supply, Transportation, Installation, Civil & Electrical works for Prefabricated/ Porta cabin at the construction site HVDC Nagpur Terminal

**1.0 Intent:**

1.1 Power Grid Corporation of India Ltd. (POWERGRID), the OWNER. intends to have Pre-fabricated/ Porta cabin site office for construction site HVDC Nagpur Terminal, Village Bhosa, Taluka-Samadrapur, Dist-Wardha, Maharashtra - 442301

**2.0 Location of Project:**

2.1 The site for the proposed work for Sub Station at Location: Village Bhosa, Taluka-Samadrapur, Dist-Wardha, Maharashtra - 442301

**3.0 Scope of Work:**

3.1 Supply, Transportation, Installation. Civil & Electrical works for ready to move Pre-Fabricated / porta cabin site office including Civil. Electrical, Finishing & Other associated works for construction site office at HVDC Nagpur Terminal.

3.2 The scope shall include the construction, facilitating measurement, carrying out Quality control tests as specified of all works of Civil and Structural works and handing over the complete ready to move Pre-Fabricated site office as per specifications.

The work to be performed under this specification shall include providing all labor, supervision, materials. storage. inventories all enabling works like scaffolding, watch and ward for the works, power, fuel, construction equipment's, water, tools and plants. supplies, transportation, all taxes and duties, all labor welfare and safety measures and all other incidental items not shown or specified. but reasonably implied or necessary for successful completion of the work including Contractor's supervision and in accordance with the drawings and specifications, inspection and testing standards, field quality control and testing as given in the Tender Documents and the complete execution of the works not mentioned here but required to complete the work.

**3.3 Scope - Civil and Structural Works:**

Supply, Transportation. Installation, all civil works, equipment and laboratory testing required for Pre-fabricated site office, interior works, steel/wood/ aluminum work, all flooring, all civil, sanitary, water supply and electrical fittings and fixtures. all electrical work, water supply/sewerage work, doors & windows. all finishes, all painting etc. and other allied works. **The DSR-2023 (Delhi Schedule of Rates published by CPWD) items for civil works referred in these technical specifications.**

**3.4 Scope - Other associated works:**

The scope shall include the works which will be required to complete the said scope of work and other envisaged and identified associated works as per site requirements and conditions.

**4.0 Classification of Items in Bill of Quantities in BID PROPOSAL SHEETS (BPS):**

4.1 For all DSR (Delhi Schedule of Rates published by CPWD) -2023 in the Bill of Quantities in Bid Proposal Sheets, the latest CPWD specification with up-to-date correction shall be followed.

4.2 For items other than DSR-2023. the relevant CPWD specification. along with the item description indicated in the BPS shall be followed. In respect of Non-Scheduled items, the unit of measurement and basis of measurement as provided in the item description of BPS shall followed. However, if unit or basis of measurements is not indicated in the tender documents, CPWD Specifications and Method of Measurement indicated in relevant IS/ CPWD specification shall be followed for such items.

4.3 The bidder shall identify and match/co-relate all the schedule and modified schedule items with the DSR-2023/CPWD Standard nomenclatures and in case of differences, if any, of minor nature such as spelling mistakes, the tender description of the items shall be deemed to be as given in the standard nomenclature of DSR-2023/CPWD. The decisions of the Owners shall be final and binding on the bidders for all such items unless clarifications regarding the same are sought before submission of tender.

4.4 For Items other than DSR-2023 items the execution and measurements of works shall be as per specifications given in this volume read in conjunction with item description in the BPS and drawings. Wherever item classification is not mentioned in BOQ, the same shall be deemed as non -DSR items.

4.5 Additional/ extra items of work taken up subsequently shall conform to CPWD specifications. In case additional/ extra items are not covered in CPWD and this volume of specifications, reference shall be made to the appropriate latest Indian/ International Standards Codes. Decision of Owner shall be the final and binding on the contractor for all such items.

**5.0 Specification for work, quality and workmanship:**

Prefabricated PUF panel system with wall of 50mm thick PUF panels made up of colour coated PPGI sheet 0.5 mm thick on each side of insulation and corrugated colour coated PUF panel 30 mm thick base and 60mm thick corrugations (minimum 30mm x 60 mm) (base layer PPGI (PPGl is Pre-painted galvanized iron and shall be used as PPGI henceforth) sheet 0.5 mm thick and upper sheet PPGI 0.5 mm thick having corrugations 60mm thick including 30mm thick base with PUF in between thoroughly; as per drawing) for roof along with all jointing Accessories, Steel Trusses etc. The PUF panels are to be fixed on raised platform

with the help of base plate, fasteners and U channels etc. as per stability and strength requirements of site. The shop drawings / manufactures drawings shall be submitted by successful bidder to POWERGRID and same are required to be approved from POWERGRID before sta1t of work.

**Design Criteria: The structure should be designed for the following parameters:**

| | | |
|---|---|---|
| a. | Density of PUF panel insulation | : - 40±2 kg/cum |
| b. | Tensile Strength of PUF | : - 4 Kg *l* Sq. Cm |
| c. | Compressive Strength (10% Deformation) | : - 2.1 Kg / Sq. Cm |
| d. | External Temperature Range | : - 0 to 55 degree Celsius |
| e. | Roof Slope | : - Ridge level should be at least 1 feet above wall level |
| f. | Thermal Conductivity 10 degree Celsius | : - Max 0.02 **W/ M.** Deg **K** |
| g. | Life of building | : - 25 years |
| h. | Water absorption | : - 0.2% volume at 100¾RH-max |
| i. | Fire Resistant | : - Fire Retardant |
| j. | Ease of construction | : - Shelters should be modular in design |
| k. | Clear Height Internal (Floor to False Ceiling) | : - 8'6" Minimum |

**Technical Specification of Component**

**5.1 Pre-fabricated Galvanized Steel Trusses:** The Truss and framework (including Foundation) design shall be prepared on STA/\D Pro model by successful bidder and same shall be submitted to POWERGRID for approval. The STAAD Pro model shall be run by successful bidder on authenticated STAAD pro software and model report (certified by structural Engineer) shall be submitted by contractor to POWERGRID. Trusses should be manufactured from Mild Steel (medium duty) hot rolled pipe section (50mm X 50mm) with provision of attachment of MS Purlin (50mm X 50mm). Further, the size of pipe / truss members may vary as per drawings prepared based on the design. All trusses shall be hot dipped Galvanized.

The design, drawings and STAAD Pro model report and submission of same (complete in all respect) shall be done by successful bidder/ contractor at his own cost and no extra amount shall be payable by POWERGRID to successful bidder/ contractor for the same.

**5.2 Framework**: The prefabricated structure frame comprises of load bearing composite wall panels held together using special CAMLOCK and silicon. Further the trusses are to be secured to the foundation with all necessary arrangements and fixtures.

The doors. windows and ventilators should be provided as per drawing and to be designed for

easy installation with help of Aluminum extrusions. The entire framework should be structurally sound and weatherproof.

**5.3 False Ceiling**: False ceiling should be provided with insulated sandwiched panels made up of two 4mm thick Fiber Reinforced Cement sheets (non-Asbestos) 42mm thick EPS core Insulation in between but without Aluminum extrusion edge. The ceiling panel should be in sizes of 1220mm X 2440mm and 1220mm X 1220mm or as required and shall be joined together with the help of ceiling joiner to ensure weatherproof sealing.

**5.4 Roof**:

a. Roof should be made of the Color coated PUF panel 30 mm thick base and 60mm thick corrugations (minimum 30mm x 60 mm) (base layer PPGI sheet 0.5 mm thick and upper sheet PPGI 0.5 mm thick having corrugations 60mm thick with including 30mm thick base with PUF in between thoroughly; as per drawing) fixed on MS purlin with all necessary arrangements, L Channel for connection with wall, fittings, screws to fix corrugated roof panel / joints. All working drawings/ manufactures drawings/ shop drawings are required to be approved from POWERGRID before sta1t of work.

b. Ridge cover should be made of PPGI 0.5 mm thick sheet matching with roof panel.

c. There should be no leakage at joints.

d. Edges of PUF panel shall be covered with 0.5mm thick PPGI sheet. At no place PUF shall be exposed to open environment.

5.5 **Flooring:** Vitrified floor tiles are to be provided in all the rooms and verandah of Site office as per DSR item no. 11.41.2. All the toilets/ wash area in site office are to be provided with ceramic floor tiles (anti skid) as per DSR item no. 11.37 and wall tiles up to height of 2.1m as per DSR item No. 8.31. Pantry of site office is to be provided with ceramic floor tiles as per DSR item no.11.37 and wall tiles as per DSR item no.8.31 up to a height of 600mm above working slab. The working slab (750mm wide) in pantry on one side along the wall is to be provided with Granite topping as per DSR item no. 8.2.2.1 /8.2.2.2.

**5.6 Doors:** The doors should be as per drawings and item no. 9.21.1 of DSR 2023 of 35mm thick flush door shutter. Door closer shall be provided as per item no-9.83 ofDSR-2023.

**5.7 Window:** The windows should be of Aluminum framed sliding glazed window as per drawing complete with all fittings, fixtures, aluminum grill (safety) and locking arrangement with double glass glazing (5.5mm (glass)+ 12mm (air gap)+ 5.5mm (glass) minimum). (shop drawing to be developed by agency and approved by Engineer-in-Charge)

**5.8 Ventilators:** The ventilators frame should be of size as per drawing with top hung hinged shutter

with 5.5mm thick glazing. Exhaust fan shall be provided as per the Technical specification of Electrical items.

**5.9 Verandah:** The verandah will be of the width as per requirement shown in the drawing. These provisions are to be made by extending the roof panel with structural support and false ceiling is to be provided.

**5.10 Galvanization:** All steel surfaces should be hot dipped galvanized as per relevant IS code and CPWD specifications.

**5.11 Gutter: Gutter** at the edges of roof shall be provided as per item no.- 12.7.1 of DSR-2023.

**5.12 Rain water Pipe:** Rain water pipe shall be provided as per item no.- 12.41.2 of DSR-2023 and it should be including necessary fitting, clamping etc. to complete the work.

**5.13 Electrification:** All electrical work to be carried out as per Electrical Technical Specifications.

**5.14 Sanitary & Plumbing:** The pre-fabricated structures should be provided with sanitary and plumbing fixtures as applicable to conventional civil buildings. The drainage system should be completely concealed in the civil flooring. All plumbing will be with cPVC pipes and fittings, which run along the external walls or the building (All work duly painted). All sanitary and plumbing work (including necessary fittings and accessories) is included in the scope as applicable for site offic.e.

The fittings required for each Toilet in **Site Office** are as follows:-

a. 1 Washbasin with 1 CP brass pillar cocks (DSR item: 17.7.4)
b. 4 CP Brass Angle valves (DSR item: 18.53.1)
c. 1 EWC' with cover and cistern (DSR item: I7.2.1)
d. 3 CP brass Bib cock (DSR item: 18.49.1)
e. I CP Brass Angle valves (DSR item: 18.53.1)
f. 1 CP brass health faucet with flexible pipe (Non Schedule item - Providing and fixing health faucet mounted on W.C. including 15mm dia flexible connection from angle valve to health faucet. complete as required. (Basic rate of health faucet=Rs.350/- each)
g. I Mirror of 600mmx 450mm (DSR item: 17.31)
h. 1 PVC water tank (1000 litrs) (DSR 18.48) on structural steel staging

The fittings required for Wash area (as per drawing) in **Site Office** are as follows:-

a. 2 Nos of Washbasin with l CP brass pillar cocks for each (DSR item.:17.7.4)
b. 2 CP Brass Angle valves (DSR item: 18.53.1)
c. I set of urinals (DSR item: 17.5.2).

The fittings required for Pantry in Site Office are as follows:-

a. I Sink with one long body bib cock and provision for another long body tap to be made (DSR item: 17. l 0.2.1 and 18.51.1)

**5.15** Septic tank (01 no.) and Soak Pit (01 no.) for Site Office are to be executed as per drawing. The location of septic tank and soak pit shall be as per site conditions.

**5.16** There will be no owner supplied material i.e. Reinforcement steel. cement etc. All supply, F&I and erection work is included in the scope of work of contractor. This provision will prevail over/ **in** place of GCC provision of Owner supplied material.

**NOTE: The quality of work will be monitored by the traceable/correlated MTC/ Vouchers submitted by the supplier in respect of material used, reference codes etc.**

**This specification has to be read in conjunction with General Conditions of Contract (GCC) and Condition of contract for civil works.**

The specifications are intended for the general descriptions of the work. quality and workmanship. The specifications are not, however, intended to cover the minute details and the work shall be executed according to the relevant CPWD Specification and in absence of CPWD specifications, the work shall be executed according to the best Public Works Department practices or to the recommendations of relevant Indian Standard/International Codes like ASTM/DfN or according to the instructions of the Engineer-in-charge.

All the materials of the project shall conform to the approved makes of materials specified in the specification and in its absence shall be of ISI make from reputed manufacturers. The procurement of various materials shall be either from the manufacturers or their main authorized dealers to ensure that no duplicate/ spurious makes are used in the works. The contractors are required to submit manufacturers' test certificates for the lots supplied at site with due endorsements by the actual dealers/ vendors/ sellers. Notwithstanding all the above, the contractor shall be held responsible for the execution of works and use of proper first class materials as per the tender specifications.

Wherever reference to CPWD/ Indian Standard Codes and practices is made, they shall be

referred to the edition/ revision of the same, issued up to 7 days prior to the date of opening of this tender.

The bidder must obtain clarification, if any, regarding the specification and all other tender documents before submission of the tender in writing with the Owner in respect of interpretation of any portions of these documents.

**All the** materials **to be used in this work should be of standard make / of ISI make or equivalent or as per specifications stipulated by CPWD. Any material to be used in the work should get approved from Engineer-in-Charge before use of the same in construction.**

**6.0 Test of Material/Workmanship:**

The extent of quality control and quality of workmanship shall be as specified in Standard field quality plan of POWERGRID (SFQP_Civil_Works_for_Site_Packages_rev_05_With Corrigendum) enclosed with Bid Documents and CPWD Specifications/ IS. Codes/ International Codes like ASTM/DIN.

All tests required for materials as desired by the Engineer-in-Charge shall be at the Contractor's cost.

All outstation visits including local visits in and around Nagpur HVDC Terminal and all around for inspection and approval of all equipment and works under this package shall be arranged by the bidders at their own cost.

**7.0 Layout and Levels:**

The layout and levels of the premises shall be made by the Contractor at his own cost from the general grid of the plot and benchmarks as approved by the Engineer-in-Charge. The Contractor shall assist with instruments. materials and men to Engineer-in-charge for checking the detailed layout and correctness of the layout and levels, at his own cost. Contractor shall be solely responsible for correctness of layout and levels. The levels and heights mentioned in the tender drawings are only tentative.

The works shall be carried out as per "Good for Construction" drawings issued after the award of the works.

**8.0 Construction Method:**

The bidder shall submit construction schedule, names of specialized agencies/ products, list of construction equipment to be deployed and write-up in a broad outline how he intends to execute the work with the best co-ordination among various sub-packages (or sub-heads). These shall form part of this Tender.

**9.0 Drawings:**

Tender drawings enclosed with this Tender Document shall form part of this specification and

supplement the requirement specified herein. The tender drawings are preliminary and meant for tender purpose only. These drawings provide a general idea about the work to be performed under the Scope of this Contact and are by no means final drawings showing the full range of work under the scope. Work has to be executed strictly according to 'Released for Construction' (or 'Good for Construction') drawings with additions, alterations and modifications made from time to time as required by the Owners. All the construction drawings shall be released by POWERGRID.

The Contractor i requested to furnish a schedule indicating their requirements of construction drawings compatible with the approved detailed construction program within 10 days of award of the work for scrutiny and approval and subsequent finalization by the

Owners. 'Good for Construction' drawings shall be issued to the contractor progressively during the execution of the contract as per actual progress achieved and requirement at site.

Any technical clarification required regarding the drawings/ specifications during the progress of works shall be obtained from the Engineer-in-charge allowing a minimum of four working days for processing. Such 'Requests For Information' shall be in approved format, which can be obtained from Engineer-in-Charge.

The actual terminal point of scope of works under this contract shall be as shown in 'Good for Construction' drawings.

**10.0 Approved vendors:**

The materials must be as per technical specifications and ISI Mark/IS Code. The Contractor shall use the material after approval of Engineer-In-Charge. If the bidder wishes to appoint any sub-vendor for any other specialized works, the same must be duly mentioned at the time of submission of tender for due approval of Owner. If at any time during the construction. the works are suspended by any approved vendor/sub-vendors. due to any reason whatsoever, the works shall be completed within the contract period by any other approved vendor as per the above lists, subject to the Owner's approval. at no additional costs, whatsoever to POWERGRID.

**11.0 Watch & Ward:**

The contractor shall be responsible for watch and ward of all the works, equipment's and various materials till complete handing over the works and all necessary arrangements to co-ordinate with watch and ward of other contractors Working simultaneously and also of the Owners shall be made to the complete satisfaction of the Engineer-in-Charge.

**12.0 General:**

12.1 The contractor shall co-operate with the representatives of the owner for due verifications of various compliances of various statutory regulations and tax compliances as and when required. All the documents of record shall be made available to the Owner and copies. wherever required shall be submitted.

12.2 Owner reserves the right to inspect all the equipment's before dispatch. However, depending

upon nature of equipment. waiver of inspection may be granted for which the contractor shall take prior approval of the Owner. This will be done on case-to-case basis.

**12.3 All the volumes of the Bid Document containing technical specification shall be read in conjunction with each other and shall be deemed to form a single document of Technical Specifications.**

**13.0 Completion Period Shall be 2 (two) months from the date of LOA**

## TECHNICAL SPECIFICATION FOR ELECTRICAL WORKS

### 1.0 SCOPE OF WORK

The scope of work includes the supply, transportation to site, erection, testing and commissioning of electrical installation work complete as required for the electrification of Porta Cabin/temporary site office/s at site HVDC Nagpur Terminal, Village Bhosa, Taluka-Samadrapur, Dist-Wardha, Maharashtra-442301 of POWERGRID shall cover following complete in all respect including accessories etc:

a. **Wiring:** Complete work including supply and fixing of conduit, wire, earthwire, accessories etc required for wiring of Light, Fan, all power Plug, exhaust fan, submains points, telephone, TV, AC, geyser etc including circuit wiring from DB to first switch board/ boxes with 2x2.5 sq.mm. insulated copper conductor and submain wiring from main supply panel to DB.
b. Providing and fixing of Switch Boards, metal boxes. prewired MCB distribution boards with MCB/MCCB/RCCB etc, all external and Internal Wiring.
c. Supply and Fixing of light fittings/fixtures/fans, bells, exhaust fans in all buildings as per requirement mentioned in this specification.
d. Providing and fixing of split type minimum 3 star rated Air conditioner at Dining, VIP room, dormitory, single and double bed room as per requirement mentioned in this specification. with all connections, complete piping, exhaust, stabilizer, industrial socket, stand etc complete in all respect.
e. Providing and fixing 25 Liter geyser for water heating at all bathrooms with all necessary connections.

f. Any other work required like earthing, cable trays, raceways etc for the complete electrification of the areas under present scope.

g. **Technical Specification: The Electrical work shall be carried out strictly in accordance with DSR Electrical (Internal).**

h. All items supplied like fixture, DB, AC, Geyser, switchgear, copper and aluminum cable etc must be as per technical specifications and ISI Mark/ IS Code. The Contractor shall use the material after approval of Engineer-In-Charge

**2.0 BILL OF QUNATITY:**

| Sl No. | Item Description | Unit | Quantity (Site Office) |
|---|---|---|---|
| 1 | Points for light/fan/ 5A power wiring | No. | 55 |
| 2 | Points for power wiring (6 pin 5/15*A* socket) and switch | No | 10 |
| 3 | TV cable socket | Nos | 02 |
| 4 | Telephone point | Nos | 05 |
| 5 | 3 Phase industrial socket with MCB (as per requirement) | Nos | 07 |
| 6 | Exhaust Fan (300 mm sweep light duty) | Nos | 02 |
| 7 | Wall Fan (300 mm sweep) | Nos | 08 |
| 8 | Ceiling Fan (1200 mm sweep) | Nos | 02 |
| 9 | LED Tube light 15/20 W | Nos | 02 |
| 10 | LED Bulb 7/10/12 W | Nos | 02 |
| II | LED down lighter (15/20/22 W) round/square (model shall be finalized in consultation with engineer-in-charge) | Nos | 20 |
| 12 | Min 3 star rated Split AC .(BEE star rated) of following rating : | | |
| (a) | 1.5 T | Nos | 05 |
| (b) | 2T | Nos | 02 |
| 13 | Geyser of capacity minimum 25 Liter.(BEE star rated) | Nos | 02 |
| 14 | Prewired Three phase vertical type Distribution board (8Way) | Nos | 01 |

**Room location/ Final location (Porta cabin or transit camp) for installation of supply items shall be finalized in consultation with Engineer-in-charge.**

**3.0 COMPLETION TESTS-On completion of the installations the following tests shall be ca1-ried out:**

- **a. Insulation resistance test.**
- **b. Polarity rest of switch.**
- **c. Earth continuity test.**